

MODEL QUESTION PAPER

TED(10) 4060
Revision 2010

Reg. No
Signature

FIFTH SEMESTER DIPLOMA EXAMINATION IN ELECTRONICS & AVIONICS

CONTROL SYSTEMS & MATLAB

(Maximum Marks : 100)

PART – A

(Answer all questions in one or two sentences. Each question carries two marks)

- I
1. State initial value theorem
 2. List various parameters of signal flow graph
 3. Define gain cross over frequency
 4. Write the syntax for array declaration in Matlab
 5. Define Transfer function of a system

PART – B

(Answer any five questions. Each question carries six marks)

- II
1. Compare open loop system and closed loop system
 2. State and explain Mason's gain formula
 3. Explain steady state error K_p for type 0 system
 4. Differentiate Absolute stability from Marginal stability with suitable diagrams
 5. Explain Routh Hurwitz criterion.
 6. Discuss Matlab functions (i) tf (ii) rlocus (iii) nyquist
 7. Explain with example 2D array declaration and its indexing in Matlab.

PART – C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT – I

- III
- | | |
|---|---|
| 1. Derive the Laplace transform for a Ramp signal | 8 |
| 2. Explain closed loop system with an example | 7 |

OR

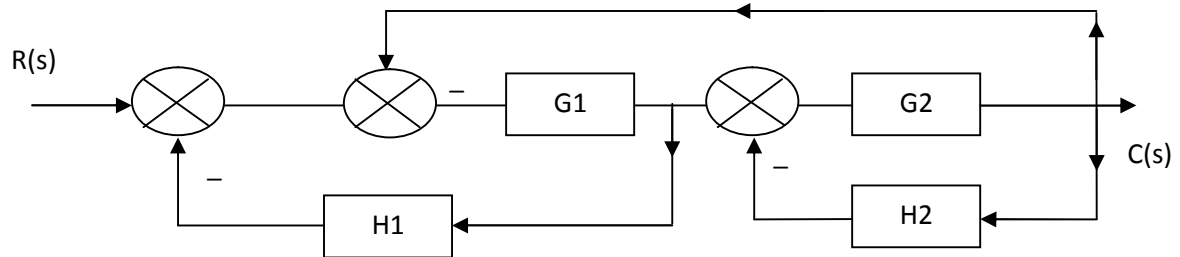
- IV
- | | |
|--|----|
| 1. Derive the transfer function of RLC series circuit. | 10 |
| 2. Find the Laplace Transform for $f(t) = e^{at}$ | 5 |

UNIT – II

V Explain block diagram reduction rules with necessary diagrams 15

OR

VI Reduce the block diagram and obtain its closed loop transfer function $C(s) / R(s)$ 15



UNIT – III

VII For $G(s)H(s) = \frac{K}{s(s+2)}$ obtain the nature of the root locus 15

OR

VIII Sketch the bode plot for the system having $G(s)H(s) = \frac{10}{s(1+0.1s)}$ 15

UNIT – IV

IX Explain the main features of Matlab with a schematic diagram. 15

OR

X Explain Mathematical operators in Matlab with examples 15